

Workshop No. 2

Kristian Ferley Mayorga Sarmiento - 20202020046

Faculty of Engineering

Computer Networks

Eng. Carlos Andrés Sierra, M.Sc.

Computer Engineering

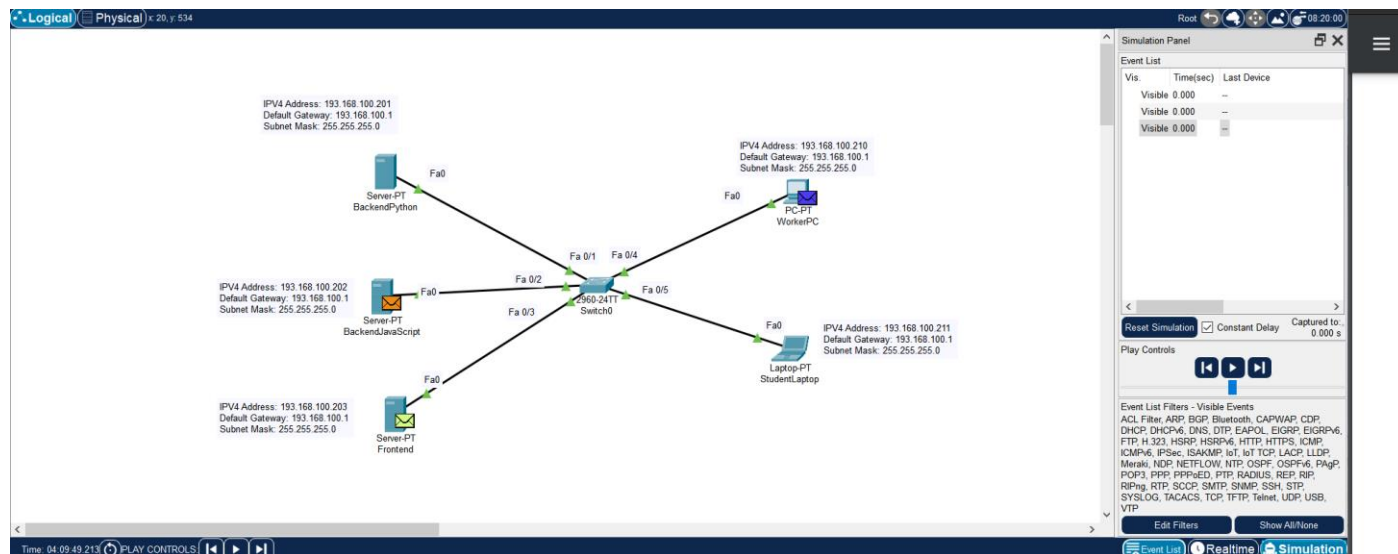


UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Universidad Distrital Francisco José de Caldas

Bogotá, Colombia

How the topology was proposed:



BackendPython Server Configuration

The screenshot shows the configuration window for the 'BackendPython' server. The 'Desktop' tab is selected, and the 'IP Configuration' section is expanded.

IP Configuration

- ☐ DHCP
- ☒ Static
- IPv4 Address: 193.168.100.201
- Subnet Mask: 255.255.255.0
- Default Gateway: 193.168.100.1
- DNS Server: 0.0.0.0

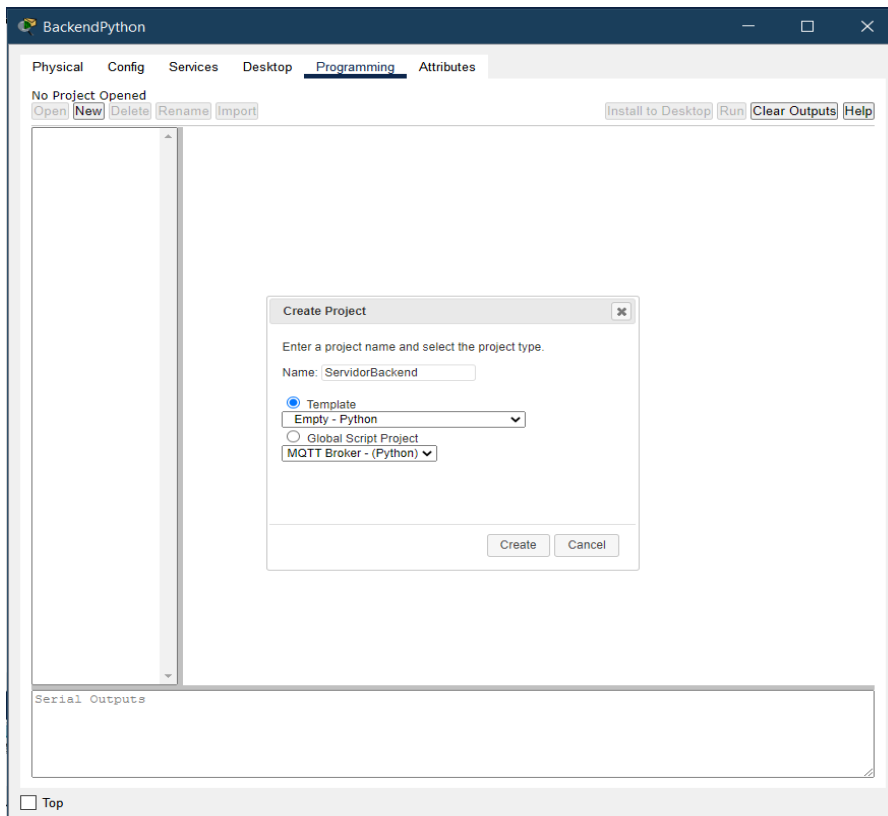
IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: [Empty field]
- Link Local Address: FE80::2E0:F9FF:FE63:80D4
- Default Gateway: [Empty field]
- DNS Server: [Empty field]

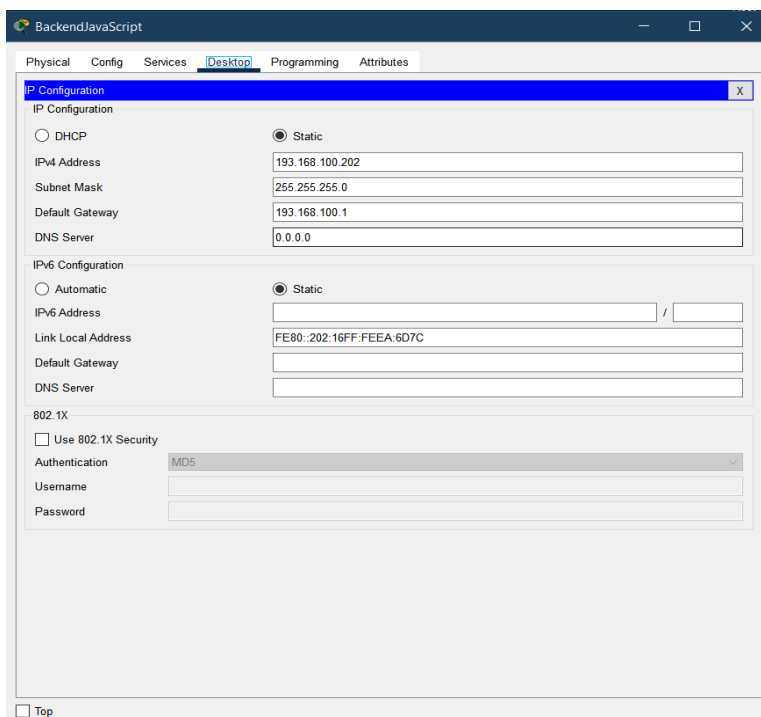
802.1X

- ☐ Use 802.1X Security
- Authentication: MD5
- Username: [Empty field]
- Password: [Empty field]

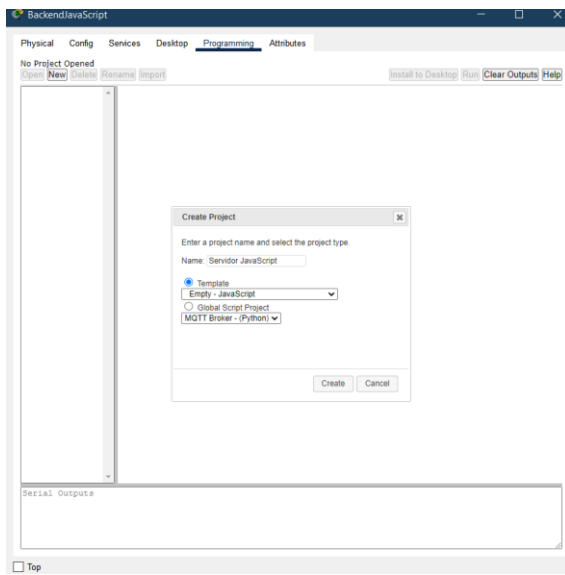
Creating the program in BackendPython



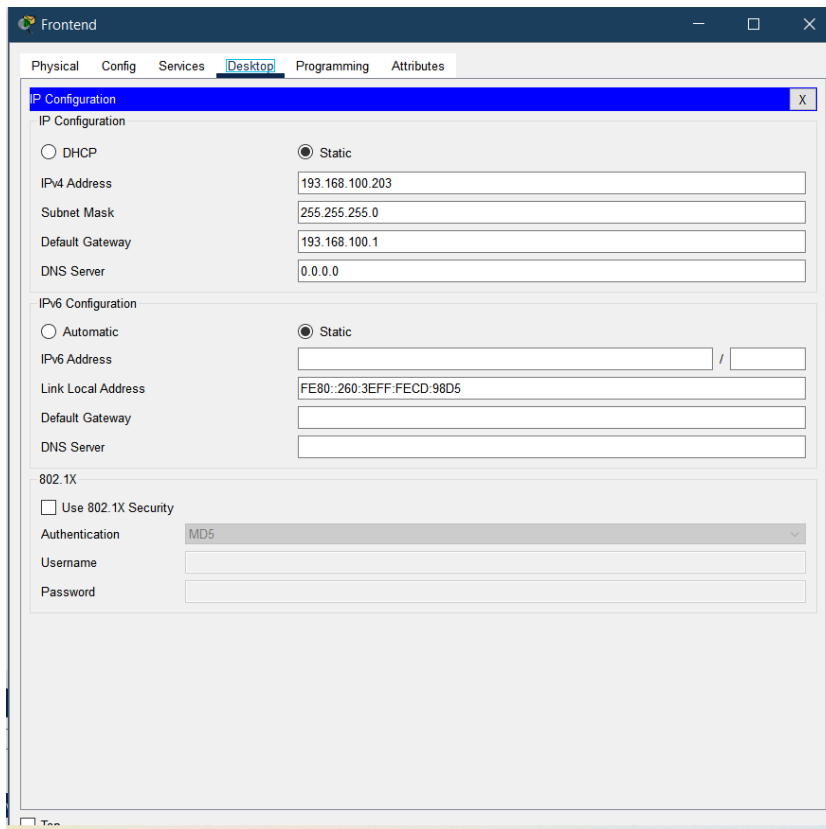
Configuring BackendJavaScript



Creating the Project



Configuring the Frontend server we have



Configure the frontend server to be a DNS

Frontend

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

OnOff

Resource Records

Name

www.udistrital.edu.co

Type

ARecord

Address

193.168.100.203

Add

Save

Remove

No.	Name	Type	Detail
-----	------	------	--------

DNS Cache

Top

Configure the IP addresses of the two computers, StudentLaptop and WorkerPC

The screenshot shows the 'StudentLaptop' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). The 'Static' configuration includes fields for 'IPv4 Address' (193.168.100.211), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (193.168.100.1), and 'DNS Server' (193.168.100.203). The 'IPv6 Configuration' section has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). The 'Static' configuration includes fields for 'IPv6 Address' (empty), 'Link Local Address' (FE80::260:3EFF:FE52:25B6), 'Default Gateway' (empty), and 'DNS Server' (empty). The '802.1X' section has a checkbox for 'Use 802.1X Security' (unchecked), a dropdown for 'Authentication' (MD5), and fields for 'Username' and 'Password' (empty). A 'Top' button is at the bottom left.

Interface	FastEthernet0
IP Configuration	
☐ DHCP ☒ Static	
IPv4 Address	193.168.100.211
Subnet Mask	255.255.255.0
Default Gateway	193.168.100.1
DNS Server	193.168.100.203
IPv6 Configuration	
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::260:3EFF:FE52:25B6
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

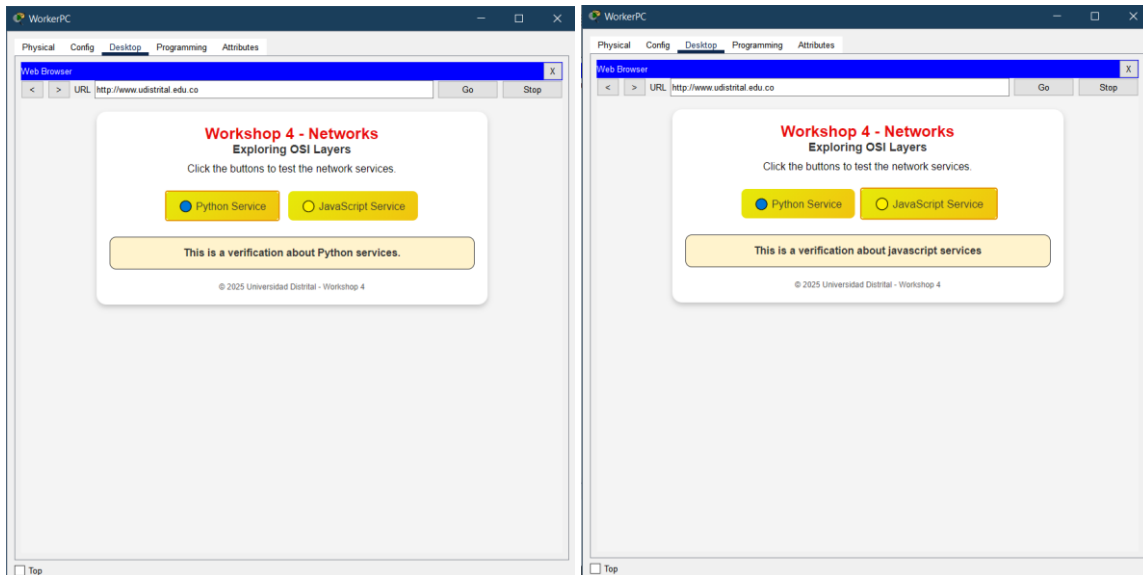
☐ Top

The screenshot shows the 'WorkerPC' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). The 'Static' configuration includes fields for 'IPv4 Address' (193.168.100.210), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (193.168.100.1), and 'DNS Server' (193.168.100.203). The 'IPv6 Configuration' section has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). The 'Static' configuration includes fields for 'IPv6 Address' (empty), 'Link Local Address' (FE80::200:CFF:FE6A:73B3), 'Default Gateway' (empty), and 'DNS Server' (empty). The '802.1X' section has a checkbox for 'Use 802.1X Security' (unchecked), a dropdown for 'Authentication' (MD5), and fields for 'Username' and 'Password' (empty). A 'Top' button is at the bottom left.

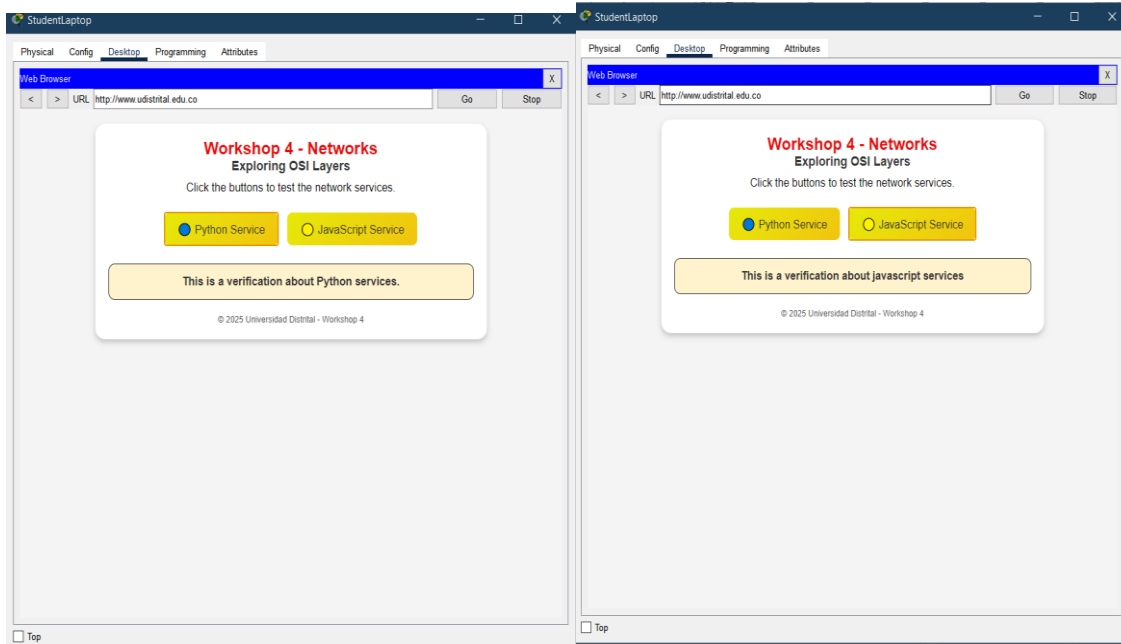
Interface	FastEthernet0
IP Configuration	
☐ DHCP ☒ Static	
IPv4 Address	193.168.100.210
Subnet Mask	255.255.255.0
Default Gateway	193.168.100.1
DNS Server	193.168.100.203
IPv6 Configuration	
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::200:CFF:FE6A:73B3
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

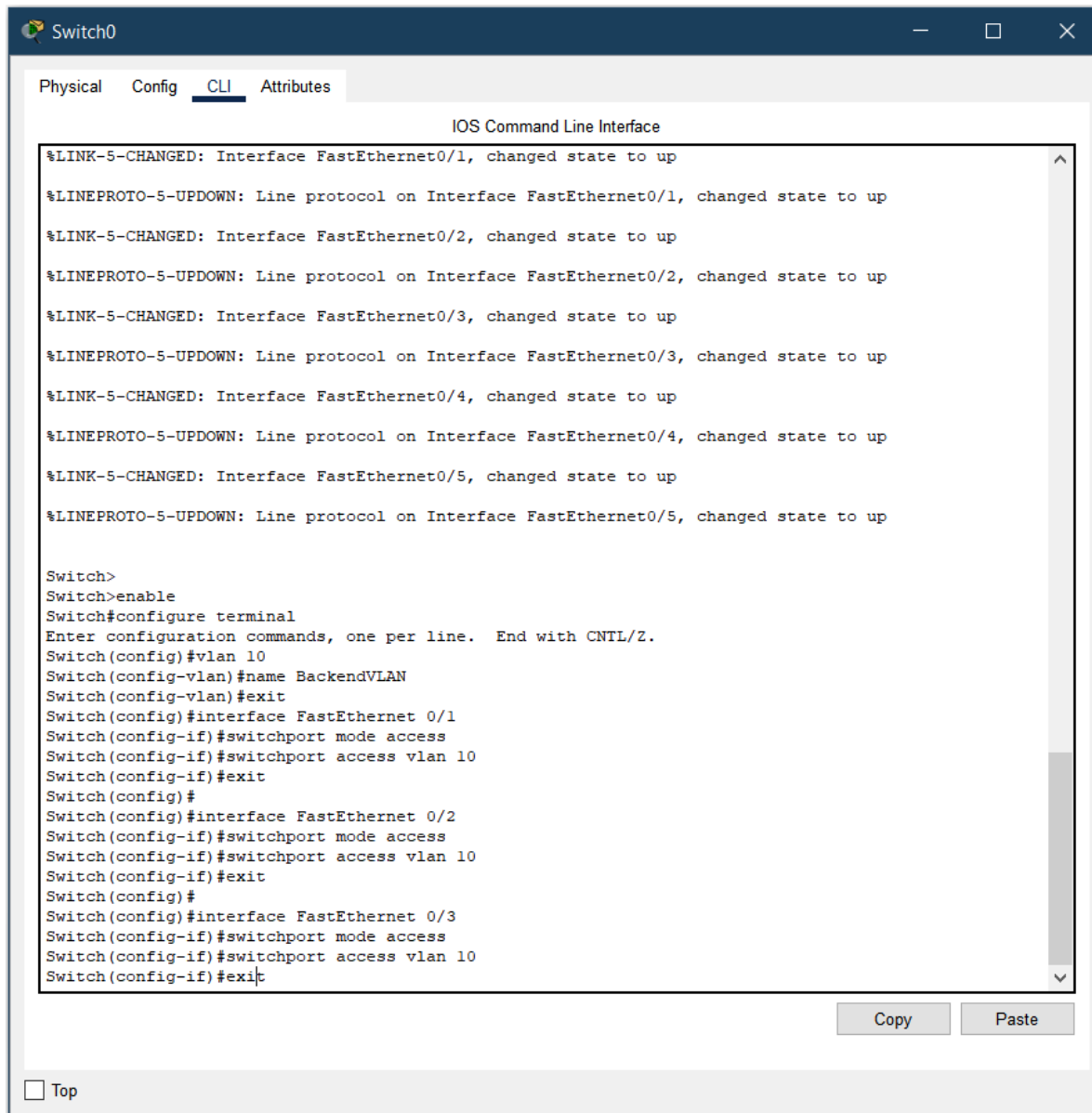
After this we proceed to verify if when placing the URL www.udistrital.edu.co it shows us the university page



With the above we observe that there is communication with our PC WorkerPC and the different servers. Checking for the other device we have:



Creating the Vlan so that the BackendPython and BackendJavaScript servers are accessible only to the FrontEnd server we have:



The screenshot shows a web-based interface for a network switch named 'Switch0'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The interface shows a series of status messages indicating that interfaces FastEthernet0/1 through FastEthernet0/5 have changed state to up. Below these, the configuration commands are entered in a terminal window. The commands create a new VLAN named 'BackendVLAN' and assign the first three interfaces (FastEthernet 0/1, 0/2, and 0/3) to this VLAN in access mode. The terminal window has a scroll bar on the right. At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button.

```
Switch0
Physical Config CLI Attributes

IOS Command Line Interface

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name BackendVLAN
Switch(config-vlan)#exit
Switch(config)#interface FastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface FastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface FastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit

Copy Paste

Top
```


Saving Vlan Configuration

```
Switch(config)#cop
Switch(config)#copy runn
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#con
Switch#copy
Switch#copy r
Switch#copy running-config sta
Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

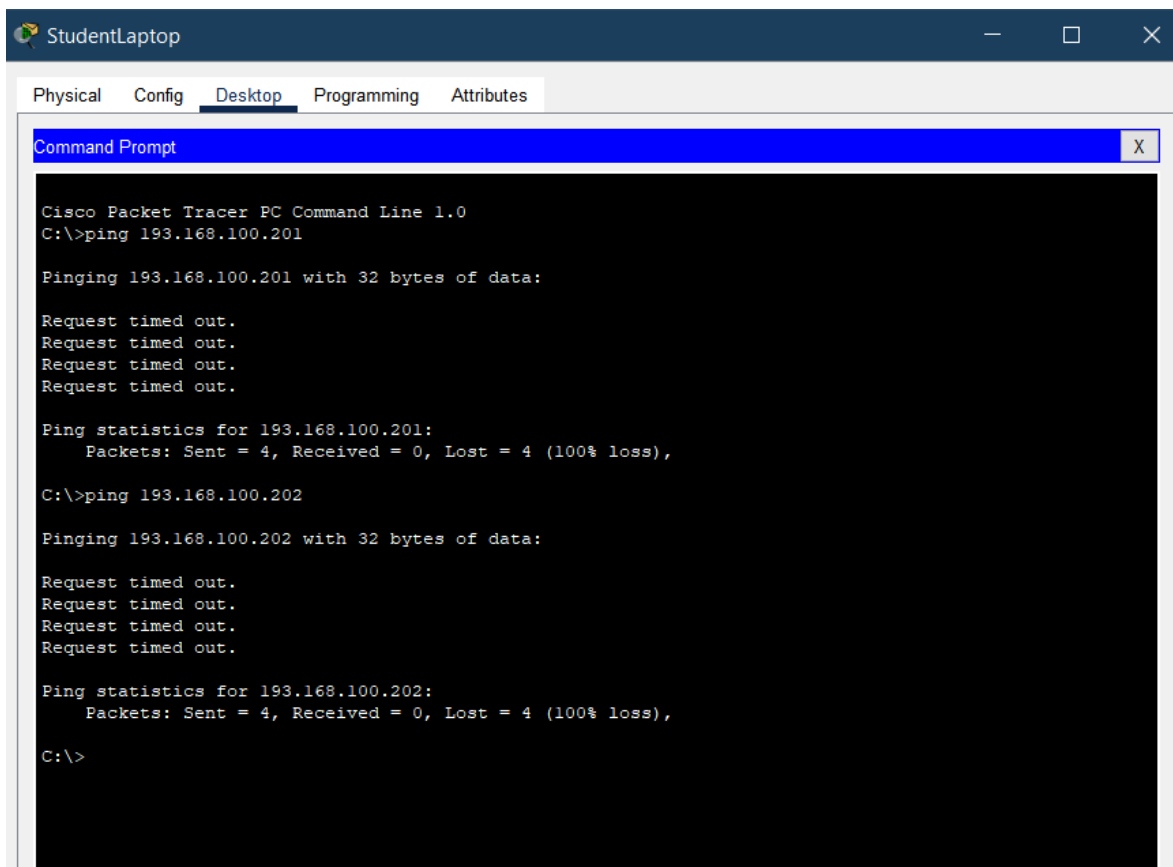
Verifying that the Vlan is created

```
[OK]
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	BackendVLAN	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

We see that the PCs are connected to the default Vlan



The screenshot shows a Cisco Packet Tracer PC Command Line window titled "StudentLaptop". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. The Command Prompt window displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 193.168.100.201

Pinging 193.168.100.201 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 193.168.100.201:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 193.168.100.202

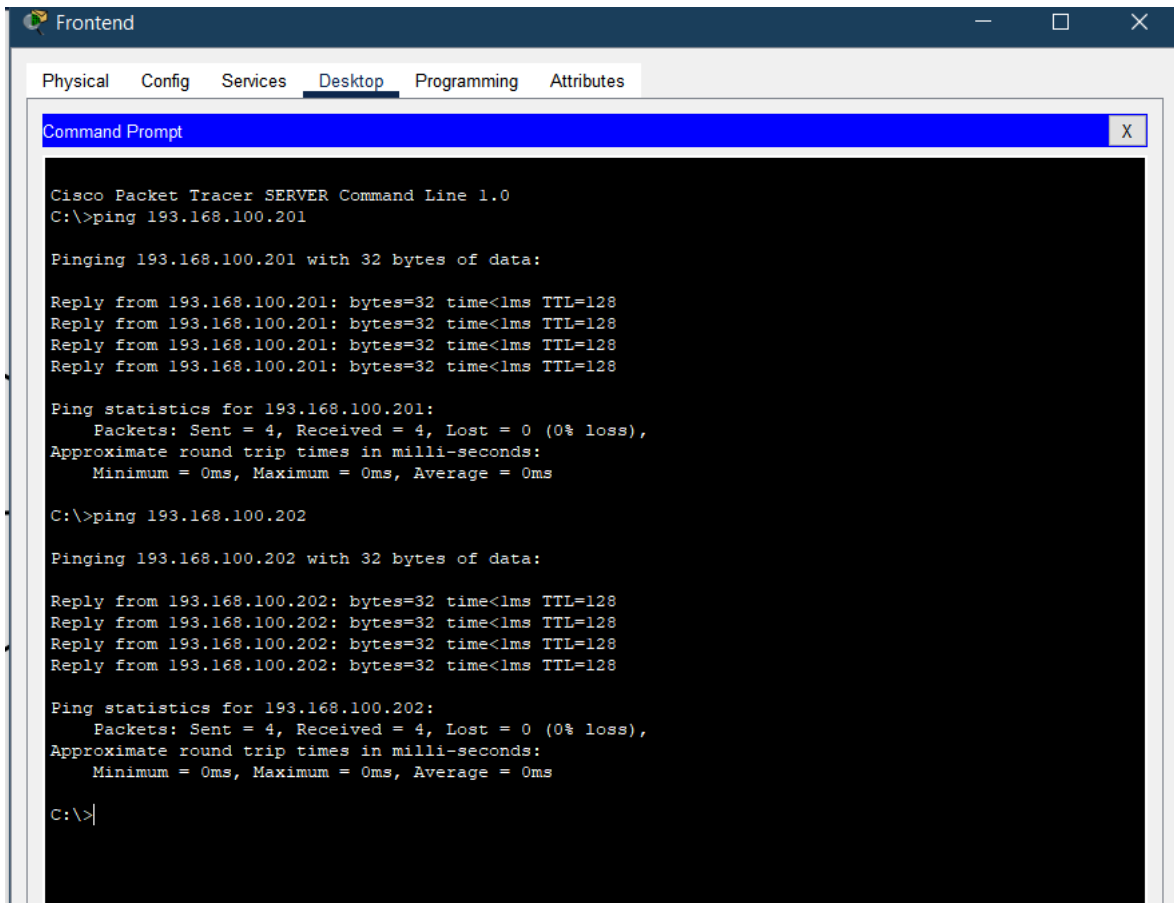
Pinging 193.168.100.202 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 193.168.100.202:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

When trying to ping both servers we see that it is not possible.



The screenshot shows a Cisco Packet Tracer Frontend window with a Command Prompt open. The Command Prompt displays the results of two ping commands. The first command is 'ping 193.168.100.201', which returns four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 128. The second command is 'ping 193.168.100.202', which also returns four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 128. The Command Prompt also shows ping statistics for both IP addresses, indicating that all packets were sent and received with 0% loss.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 193.168.100.201

Pinging 193.168.100.201 with 32 bytes of data:

Reply from 193.168.100.201: bytes=32 time<1ms TTL=128
Reply from 193.168.100.201: bytes=32 time<1ms TTL=128
Reply from 193.168.100.201: bytes=32 time<1ms TTL=128
Reply from 193.168.100.201: bytes=32 time<1ms TTL=128

Ping statistics for 193.168.100.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 193.168.100.202

Pinging 193.168.100.202 with 32 bytes of data:

Reply from 193.168.100.202: bytes=32 time<1ms TTL=128
Reply from 193.168.100.202: bytes=32 time<1ms TTL=128
Reply from 193.168.100.202: bytes=32 time<1ms TTL=128
Reply from 193.168.100.202: bytes=32 time<1ms TTL=128

Ping statistics for 193.168.100.202:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

When pinging from the FrontEnd Server we see that there is a connection as requested in practice.